A CONDITIONAL EQUILIBRIUM CRITERION FOR COMPACT ANCIENT NAVIER–STOKES DYNAMICS

GUILLEM DURAN BALLESTER

Abstract. We prove a conditional equilibrium criterion for compact ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables. The assumption is formulated on a compact time-translation invariant hull: there is a continuous Lyapunov functional whose dissipation is nonnegative and vanishes precisely on stationary trajectories. Under this assumption, every invariant probability measure is supported on stationary solutions. If the compact hull is minimal and uniformly bounded in $L^3(\mathbb{R}^3)$, the stationary self-similar theorem of Necas–Ruzicka–Sverak forces the hull to consist only of the zero trajectory. We then record the resulting conditional assembly: a normalized collection has no nonzero ancient solutions provided its minimal elements generate compact nonzero hulls satisfying the Lyapunov hypothesis. The construction of such a Lyapunov functional is not asserted here; it is isolated as the remaining analytic input.

1. Introduction

We consider the incompressible Navier–Stokes equations in backward self-similar variables,

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \operatorname{div} V = 0, \tag{1.1}$$

on $\mathbb{R}^3\times\mathbb{R}$. The equation is autonomous in τ. If V is a solution and $s\in\mathbb{R}$, then

$$V^s(y,\tau) = V(y,\tau+s)$$

is again a solution.

The compactness and invariant-measure construction for time translates was proved in the preceding compact-dynamics paper [5]. That construction yields a stationary statistical identity, but it does not imply that the underlying trajectories are stationary: the barycenter of an invariant measure carries a quadratic covariance term. The additional input considered here is an equilibrium principle excluding recurrent nonstationary compact motion. This paper records a precise conditional form of that principle.

The hypothesis used below is explicit. We assume the existence of a continuous Lyapunov functional on the compact invariant hull, with an exact dissipation identity and with zero dissipation precisely at stationary trajectories. This hypothesis implies that invariant measures are supported on equilibria. Uniform L^3 control then places stationary elements within the scope of the theorem of Necas–Ruzicka–Sverak [6]. Thus the paper is a conditional assembly theorem, not an unconditional proof of equilibrium rigidity for all compact ancient Navier–Stokes dynamics.

2020 *Mathematics Subject Classification.* 35Q30, 35B40, 35B41, 35B44, 37B25, 76D05.

Key words and phrases. Navier–Stokes equations, ancient solutions, invariant measures, self-similar variables, Type I singularities, Liouville theorems.

1.1. **Role of the hypothesis.** The strict Lyapunov hypothesis is the only input in this paper not already provided by the preceding compactness, minimality, and stationary-rigidity arguments. Compactness gives invariant measures, and stationary L^3 rigidity gives zero stationary profiles. The remaining point is the equilibrium-support assertion: an invariant measure on a compact nonlinear Navier–Stokes hull must be supported on stationary trajectories. The functional E and dissipation D in Theorem 3.1 encode precisely that assertion.

2. Compact ancient dynamics

We use the same solution class, topology, and compact-hull conventions as in the compact-dynamics paper [5, Sections 2–5]. They are recalled here only to fix notation for the present argument. The locally smooth topology is the standard Fréchet topology of uniform convergence of all space-time derivatives on compact subsets. It is induced by the countable family of seminorms

$$\|U\|_{m,N} = \max_{|\alpha|+j\le m} \sup_{\substack{|y|\le N\\ |\tau|\le N}} |\partial_y^\alpha \partial_\tau^j U(y,\tau)|, \qquad m, N \in \mathbb{N}.$$

The metric

$$d(U,V) = \sum_{m,N=1}^{\infty} 2^{-(m+N)} \frac{\|U-V\|_{m,N}}{1+\|U-V\|_{m,N}}$$

induces this topology. Consequently each compact hull considered here is a compact metric space.

All probability measures below are Borel probability measures on the relevant compact metric hull K, and $C(K)$ denotes the real-valued continuous functions on K. Weak convergence of probability measures is understood against test functions in $C(K)$. Since K is compact and metric, every Borel probability measure on K is Radon; in particular its closed support has full measure. A probability measure μ is invariant if

$$\int_K F(\Theta_t W)\,d\mu(W) = \int_K F(W)\,d\mu(W) \qquad \text{for every } F\in C(K) \text{ and every } t\in\mathbb{R}.$$

If $B \subset K$ is Borel, the phrase "μ is supported on B" means $\mu(K\setminus B)=0$. The closed support $\operatorname{supp}\mu$ is the smallest closed subset of K with full μ-measure, equivalently the set of points whose every open neighborhood has positive μ-measure.

After applying the whole-space Helmholtz–Leray projection $\mathbb{P}$, equation (1.1) becomes

$$\partial_\tau V = \Delta V - \frac{1}{2} y\cdot\nabla V - \frac{1}{2}V - \mathbb{P}\operatorname{div}(V\otimes V). \tag{2.1}$$

Let $S(a)$, $a>0$, denote the semigroup generated by $\Delta - \frac{1}{2}y\cdot\nabla - \frac{1}{2}$. The pressure reconstruction and the passage between the projected mild equation and the local classical Navier–Stokes equation are those proved in the lemma "Local smoothing and pressure reconstruction" of [5].

The mild formulation is used only to specify the ancient solution class and to make the time-translation action intrinsic. The arguments below use the locally smooth representatives in the compact hull and distributional identities obtained from (1.1). The distribution $\mathbb{P}\operatorname{div}(V\otimes V)$ is interpreted as in [5, Definition "Bounded mild ancient solution"], by duality against smooth test fields and as a tempered-distribution Fourier multiplier on each time slice. No boundedness of the Riesz transforms on L^∞ is used in this paper.

Definition 2.1 (Bounded mild ancient solution)**.** A vector field $V:\mathbb{R}^3\times\mathbb{R}\to\mathbb{R}^3$ is a bounded mild ancient solution of (1.1) if

$$V\in L^\infty(\mathbb{R}^3\times\mathbb{R}), \qquad \operatorname{div} V = 0$$

in distributions, and if for every $\sigma < \tau$

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s)\mathbb{P}\,\mathrm{div}(V \otimes V)(s)\,ds \tag{2.2}$$

as an identity in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

Definition 2.2 (Compact invariant hull)**.** Let K be a nonempty compact set of bounded mild ancient solutions, with compactness understood in the locally smooth

$$C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$$

topology. For $t \in \mathbb{R}$, write

$$\Theta_t W = W^t, \qquad W^t(y, \tau) = W(y, \tau + t).$$

The set K is invariant if $\Theta_t K = K$ for every $t \in \mathbb{R}$. It is minimal if it has no proper nonempty compact invariant subset.

Lemma 2.3 (Continuity of time translation)**.** *For each $t \in \mathbb{R}$, the map $\Theta_t : K \to K$ is a homeomorphism. Moreover, for every $W \in K$, the orbit map*

$$\mathbb{R} \ni t \mapsto \Theta_t W \in K$$

is continuous, and the action map

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is continuous.

Proof. This is the lemma "Continuity of the restricted translation action" in [5]. The result is stated for the same locally smooth topology and the same compact invariant sets used here. It proves continuity of each time-translation map, joint continuity of the action, and the identity $\Theta_t^{-1} = \Theta_{-t}$, hence the asserted homeomorphism property. □

Definition 2.4 (Stationary trajectory)**.** An element $W \in K$ is stationary if $\Theta_t W = W$ for every $t \in \mathbb{R}$. For a time-independent trajectory $W(y, \tau) = w(y)$, the corresponding stationary self-similar profile equation is

$$(w \cdot \nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \qquad \mathrm{div}\, w = 0. \tag{2.3}$$

Lemma 2.5 (Distributions with zero gradient)**.** *Let $\Omega \subset \mathbb{R}^3$ be connected and open. If $q \in \mathcal{D}'(\Omega)$ satisfies $\nabla q = 0$ in $\mathcal{D}'(\Omega)$, then q is constant in Ω, in the sense that there exists $c \in \mathbb{R}$ with $q = c$ in $\mathcal{D}'(\Omega)$.*

Proof. Let ρ_ε be a Friedrichs mollifier. If $\Omega' \Subset \Omega$ and $\varepsilon < \mathrm{dist}(\Omega', \partial\Omega)$, then

$$q_\varepsilon(x) = \langle q(y), \rho_\varepsilon(x - y)\rangle$$

is well-defined and smooth for $x \in \Omega'$. Moreover,

$$\nabla q_\varepsilon = (\nabla q) * \rho_\varepsilon = 0 \qquad \text{on } \Omega'.$$

Thus q_ε is constant on every connected component of Ω'. In particular, if $B \Subset \Omega$ is a ball, then q_ε is constant on B for all sufficiently small ε. To see that the limit is also constant, let $\varphi \in C_c^\infty(B)$ satisfy $\int_B \varphi\,dx = 0$. Since q_ε is constant on B,

$$\langle q_\varepsilon, \varphi\rangle = 0.$$

Passing to the limit gives $\langle q, \varphi \rangle = 0$ for every such φ. Choose $\eta \in C_c^\infty(B)$ with $\int_B \eta \, dx = 1$, and set $c = \langle q, \eta \rangle$. For arbitrary $\psi \in C_c^\infty(B)$, the test function $\psi - (\int_B \psi \, dx)\eta$ has integral zero; hence

$$\langle q, \psi \rangle = c \int_B \psi \, dx.$$

Thus the restriction of q to B is the constant distribution c.

If $B_1, B_2 \Subset \Omega$ are balls with $B_1 \cap B_2 \neq \varnothing$, the constants obtained on B_1 and B_2 agree, because both represent the restriction of q to $B_1 \cap B_2$. It remains only to justify that this propagates through all of Ω. Fix a ball $B_0 \Subset \Omega$, and let U be the union of all balls $B \Subset \Omega$ that can be joined to B_0 by a finite chain of compactly contained balls with consecutive nonempty intersections. The set U is open by construction. It is also closed relative to Ω: if $x_j \in U$ and $x_j \to x \in \Omega$, choose a ball $B_x \Subset \Omega$ containing x; then $x_j \in B_x$ for large j, so the chain reaching a ball containing x_j, followed by B_x, reaches B_x. Since Ω is connected and U is nonempty, we have $U = \Omega$. Hence the local constants agree throughout Ω. □

Lemma 2.6 (Stationary elements and the profile equation)**.** *Let $W \in K$. Then W is stationary if and only if there is a smooth divergence-free vector field $w : \mathbb{R}^3 \to \mathbb{R}^3$ such that*

$$W(y, \tau) = w(y) \qquad \textit{for all } (y, \tau) \in \mathbb{R}^3 \times \mathbb{R}.$$

For every such w, there is a scalar distribution p, unique modulo additive constants, such that w solves (2.3) in $\mathcal{D}'(\mathbb{R}^3)$.

Proof. If W is stationary, then $\Theta_t W = W$ as elements of the locally smooth function space for every $t \in \mathbb{R}$. Equality in this Hausdorff function space is equality of the smooth representatives; hence

$$W(y, \tau + t) = W(y, \tau) \qquad \text{for all } y, \tau, t.$$

Taking $\tau = 0$ and then writing t as an arbitrary time variable gives $W(y, t) = W(y, 0)$. Thus W is independent of time. Conversely, if $W(y, \tau) = w(y)$, then $W(y, \tau + t) = W(y, \tau)$ for every t, so $\Theta_t W = W$ for every t.

Set $w(y) = W(y, 0)$. The local smoothness of W gives $w \in C^\infty(\mathbb{R}^3; \mathbb{R}^3)$, and the distributional condition $\operatorname{div} W = 0$ gives $\operatorname{div} w = 0$. By the lemma "Local smoothing and pressure reconstruction" in [5], every bounded mild ancient solution in the compact hull has, on each compact space-time cylinder, a smooth local pressure for which (1.1) holds pointwise. For each integer $N \geq 1$, apply that result on the cylinder $B_N \times (-1, 1)$. There is a smooth pressure $P_N(y, \tau)$ on this cylinder satisfying (1.1) with $V = W$. Since the present W is independent of τ, the vector field

$$\Delta w - \frac{1}{2}(w + y \cdot \nabla w) - (w \cdot \nabla) w$$

is independent of τ, and therefore $\nabla_y P_N(y, \tau)$ is independent of τ on $B_N \times (-1, 1)$. Setting $p_N(y) = P_N(y, 0)$, the local identity reduces on B_N to

$$(w \cdot \nabla) w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \qquad \operatorname{div} w = 0.$$

On overlaps the gradients of two such local pressures agree, because the remaining terms in the displayed equation are global functions of w. Set $c_1 = 0$. For each integer $R \geq 2$, apply Theorem 2.5 on B_{R-1} to the distribution $p_R - p_{R-1} - c_{R-1}$. Its gradient vanishes on B_{R-1}, because both ∇p_R and ∇p_{R-1} are equal there to

$$\Delta w - \frac{1}{2}(w + y \cdot \nabla w) - (w \cdot \nabla) w.$$

Therefore the distribution is constant on B_{R-1}. Choose c_R to be the negative of that constant. Then

$$p_R + c_R = p_{R-1} + c_{R-1} \qquad \text{in } \mathcal{D}'(B_{R-1}).$$

Define the global distribution p as follows. If $\varphi \in C_c^\infty(\mathbb{R}^3)$, choose an integer $R \geq 1$ such that $\operatorname{supp}\varphi \subset B_R$, and set

$$\langle p, \varphi\rangle = \langle p_R + c_R, \varphi\rangle.$$

This definition is independent of the chosen sufficiently large R, because the preceding compatibility relation iterates: if $S > R$ are integers, then

$$p_S + c_S = p_R + c_R \qquad \text{in } \mathcal{D}'(B_R).$$

Indeed, one applies the preceding compatibility identity successively on $B_{S-1}, B_{S-2}, \dots, B_R$, and then restricts the resulting distributional identity to B_R. Hence p is a well-defined scalar distribution on $\mathbb{R}^3$, and its gradient agrees with the patched local pressure gradients.

Let $\Phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$. Choose an integer R so large that $\operatorname{supp}\Phi \subset B_R$. Testing the local identity on B_R against Φ, and using the definition of p on B_R, gives

$$\left\langle (w\cdot\nabla)w + \nabla p - \Delta w + \frac{1}{2}(w + y\cdot\nabla w), \Phi \right\rangle = 0.$$

Since this holds for every compactly supported smooth vector field Φ, (2.3) holds in $\mathcal{D}'(\mathbb{R}^3)$. If p and $\tilde p$ both have the required gradient, then $\nabla(p - \tilde p) = 0$ in $\mathcal{D}'(\mathbb{R}^3)$, so Theorem 2.5 applied on the connected set $\mathbb{R}^3$ gives $p - \tilde p$ constant. $\square$

Theorem 2.7 (Invariant measures on compact hulls)**.** *Let K be a compact invariant hull and let $W_0 \in K$. For $T > 0$, set*

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds.$$

Equivalently, μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$ under the continuous orbit map $s \mapsto \Theta_s W_0$; hence μ_T is a Borel probability measure on K. Every sequence $T_n \to \infty$ has a subsequence for which μ_{T_n} converges weakly to an invariant Borel probability measure μ on K. If K is minimal, then the support of every invariant Borel probability measure on K is all of K.

Proof. This is the Krylov–Bogolyubov averages theorem proved in [5, Theorem "Krylov–Bogolyubov averages"]. The hypotheses match because Theorem 2.2 uses the same compact locally smooth hull and time-translation flow. That theorem gives subsequential weak limits of the orbit averages, invariance under every Θ_t, and the full-support conclusion in the minimal case. $\square$

3. The equilibrium hypothesis

Hypothesis 3.1 (Strict Lyapunov functional)**.** Let K be a compact invariant hull. Assume there are continuous functions

$$E : K \to \mathbb{R}, \qquad D : K \to [0, \infty)$$

such that, for every $W \in K$ and every $t \geq 0$,

$$E(\Theta_t W) - E(W) = -\int_0^t D(\Theta_s W)\, ds. \tag{3.1}$$

The integral is the ordinary Lebesgue integral of the continuous function $[0,t] \ni s \mapsto D(\Theta_s W)$. Thus (3.1) is an integral identity along each orbit; no differentiability of $E(\Theta_t W)$ in t is assumed. Assume also that

$$D(W) = 0 \quad \text{if and only if} \quad W \text{ is stationary.} \tag{3.2}$$

Remark 3.2. The paper proves the consequences of Theorem 3.1. It does not claim that such a functional has been constructed for the full Navier–Stokes dynamics. Constructing it, or replacing it by an equivalent rigidity principle, is the remaining equilibrium problem.

Lemma 3.3 (Invariant measures annihilate dissipation)**.** *Let K be a compact invariant hull satisfying Theorem 3.1. If μ is an invariant Borel probability measure on K, then*

$$\int_K D(W)\, d\mu(W) = 0.$$

Proof. Fix $t > 0$. The functions E and D are continuous on compact K, so they are bounded and Borel measurable. By Theorem 2.3, $(s, W) \mapsto D(\Theta_s W)$ is continuous on $[0,t] \times K$. In particular it is Borel measurable and bounded on the compact product $[0,t] \times K$. With respect to the finite product measure $ds \otimes d\mu(W)$, all integrals below are finite, and Fubini's theorem applies.

Integrating (3.1) with respect to μ, and using invariance of μ with the continuous test function E, gives

$$0 = \int_K E(\Theta_t W)\, d\mu(W) - \int_K E(W)\, d\mu(W) = -\int_K \int_0^t D(\Theta_s W)\, ds\, d\mu(W).$$

By Fubini's theorem and invariance again, now with the continuous test function D, for each fixed $s \in [0,t]$,

$$\int_K \int_0^t D(\Theta_s W)\, ds\, d\mu(W) = \int_0^t \int_K D(\Theta_s W)\, d\mu(W)\, ds = t \int_K D(W)\, d\mu(W).$$

Combining the two displayed identities gives

$$t \int_K D(W)\, d\mu(W) = 0.$$

Since $t > 0$, the claim follows. □

Lemma 3.4 (Zero dissipation gives stationary support)**.** *Let $D : K \to [0,\infty)$ be continuous and let μ be a Borel probability measure on K. If*

$$\int_K D(W)\, d\mu(W) = 0,$$

then μ is supported on the closed set $\{D = 0\}$.

Proof. For each $\varepsilon > 0$, the set

$$K_\varepsilon = \{W \in K :\ D(W) \geq \varepsilon\}$$

is closed, hence Borel. Since $D \geq \varepsilon$ on K_ε,

$$0 = \int_K D\, d\mu \geq \varepsilon\, \mu(K_\varepsilon).$$

Thus $\mu(K_\varepsilon) = 0$ for every $\varepsilon > 0$. Since D is continuous, the zero set $\{D = 0\}$ is closed. Its complement is the countable union

$$K \setminus \{D = 0\} = \bigcup_{n=1}^{\infty} K_{1/n}.$$

Indeed, if $D(W) > 0$, then $1/n \leq D(W)$ for all sufficiently large n. Hence $K \setminus \{D = 0\}$ has μ-measure zero. $\square$

Theorem 3.5 (Equilibrium support)**.** *Let K be a compact invariant hull satisfying Theorem 3.1. Then every invariant Borel probability measure on K is supported on stationary trajectories.*

Proof. Let μ be an invariant Borel probability measure on K. By Theorem 3.3,

$$\int_K D(W)\, d\mu(W) = 0.$$

By Theorem 3.4, μ is supported on $\{D = 0\}$. The zero set of D is precisely the set of stationary trajectories by (3.2). Hence every invariant measure is supported on stationary trajectories. $\square$

Corollary 3.6 (Minimal compact hulls are stationary)**.** *Let K be a minimal compact invariant hull satisfying Theorem 3.1. Then every element of K is stationary. Consequently K consists of a single stationary trajectory.*

Proof. Choose $W_0 \in K$ and a sequence $T_n \to \infty$. By Theorem 2.7, after passing to a subsequence the averages

$$\mu_{T_n} = \frac{1}{T_n} \int_0^{T_n} \delta_{\Theta_s W_0}\, ds$$

converge weakly to an invariant Borel probability measure μ on K. Since K is minimal, the support assertion in Theorem 2.7 yields $\operatorname{supp} \mu = K$.

By Theorem 3.5, μ is supported on stationary trajectories. By Theorem 3.1, the set of stationary trajectories is $\{D = 0\}$, and this set is closed because D is continuous. We claim that $\operatorname{supp} \mu \subset \{D = 0\}$. Indeed, if $W \in \operatorname{supp} \mu \setminus \{D = 0\}$, then the open set

$$U = K \setminus \{D = 0\}$$

is an open neighborhood of W. By the definition of support, $\mu(U) > 0$. This contradicts $\mu(K \setminus \{D = 0\}) = 0$. Therefore

$$K = \operatorname{supp} \mu \subset \{W \in K : \ W \text{ is stationary}\}.$$

Hence every element of K is stationary.

If $W \in K$ is stationary, then the singleton $\{W\}$ is compact and nonempty. It is invariant because $\Theta_t W = W$ for every t. Since K has no proper nonempty compact invariant subset, minimality gives $K = \{W\}$. $\square$

4. Stationary self-similar rigidity

Theorem 4.1 (Stationary self-similar theorem)**.** *Let $w \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ solve (2.3) in distributions for some scalar distribution p, with $\operatorname{div} w = 0$. Then $w \equiv 0$.*

Proof. This is the stationary self-similar Liouville theorem of Necas–Ruzicka–Sverak [6], recalled in this series as the stationary self-similar rigidity theorem in [3]. Their theorem is stated for weak $L^3(\mathbb{R}^3)$ solutions of the stationary Leray self-similar system; the smooth distributional formulation above is a special case. $\square$

Theorem 4.2 (Rigidity of compact tight minimal dynamics)**.** *Let K be a minimal compact invariant hull satisfying Theorem 3.1. Assume that there is $L < \infty$ such that*

$$\sup_{W \in K} \sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L.$$

Then every element of K is identically zero.

Proof. By Theorem 3.6, $K = \{W\}$ for a stationary trajectory. By Theorem 2.6, there is a smooth divergence-free profile w such that $W(y,\tau) = w(y)$, and w solves (2.3) in the sense of distributions for some pressure. Since $W \in K$, the assumed bound gives

$$\|w\|_{L^3(\mathbb{R}^3)} = \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L \qquad \text{for every } \tau \in \mathbb{R}.$$

Thus $w \in L^3(\mathbb{R}^3)$. The hypotheses of Theorem 4.1 are satisfied, and hence $w \equiv 0$. Therefore W is the identically zero trajectory, and every element of K is identically zero. □

5. Conditional assembly

The preceding theorem is a compact-dynamical rigidity statement. We now record the precise form in which it combines with the minimal-element reduction developed in [4]. The reduction itself is not reproved here; it is isolated as the hypothesis below so that this paper only supplies the compact-equilibrium implication.

Hypothesis 5.1 (Minimal compact reduction)**.** Let A be a collection of bounded mild ancient solutions of (1.1). Assume the following.

(1) If A is nonempty, then A contains a nonzero element V_* of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm among all elements of A.
(2) For every such minimal element V_*, the closure of $\{V_*^s : \ s \in \mathbb{R}\}$ contains a minimal compact invariant hull K which contains a nonzero trajectory.
(3) Every such K satisfies Theorem 3.1.
(4) For every such compact invariant hull K, there is $L_K < \infty$ such that every $W \in K$ satisfies

$$\sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L_K.$$

Remark 5.2. This hypothesis is formulated for normalized collections, such as the fixed nonvanishing classes obtained in the minimal-element reduction. In particular, if A is nonempty, then the first item excludes the identically zero trajectory from A.

Theorem 5.3 (Conditional ancient Liouville theorem)**.** *If A satisfies Theorem 5.1, then A is empty.*

Proof. Suppose that A is nonempty. By the first part of Theorem 5.1, choose a nonzero minimal element $V_* \in A$. By the second part, the time-translation hull of V_* contains a minimal compact invariant hull K containing a nonzero trajectory. By the third and fourth parts, K satisfies the hypotheses of Theorem 4.2. Therefore every element of K is identically zero. This contradicts the second part of Theorem 5.1, which asserts that the same K contains a nonzero trajectory. The assumption that A is nonempty is therefore false. □

Corollary 5.4 (Conditional Type I exclusion)**.** *Assume that a putative Type I singularity in a given class produces, through the Type I blow-up procedure cited in [2], a nonzero bounded mild ancient solution belonging to a collection A satisfying Theorem 5.1. Then no Type I singularity occurs in that class.*

Proof. The nonzero centered ancient limit is the Type I blow-up output used in [2]; the local concentration/nonvanishing input underlying the surrounding singularity dichotomy is recorded in [1]. Thus the blow-up construction would give a nonzero element of A, while Theorem 5.3 gives $A = \emptyset$. Since a set cannot both be empty and contain the produced nonzero ancient solution, this contradiction excludes the singularity. $\square$

6. Conclusion

The conditional structure is as follows. Compactness and invariant measures reduce compact ancient dynamics to invariant probability measures. A strict Lyapunov identity forces all such measures to be supported on stationary trajectories. Uniform L^3 control then invokes the stationary self-similar theorem and eliminates every nonzero minimal compact hull. Thus the remaining analytic task is to prove the strict Lyapunov hypothesis, or an equivalent equilibrium-support theorem, for compact hulls generated by minimal ancient Navier–Stokes solutions.

References

[1] G. Cordero, *Local CKN Concentration and the Type I/Type II Dichotomy for Navier–Stokes*, manuscript in this series, `docs/source/first_nodes/paper0_local_concentration_entry_typeII.tex`.

[2] G. Duran Ballester, *Type I Blow-up Limits for the Three-dimensional Navier–Stokes Equations and the Centered Ancient Equation*, manuscript in this series, `docs/source/type_1/paper1_typeI_blowup_limits_centered_ancient.tex`.

[3] G. Duran Ballester, *Uniformly Tight Ancient Solutions of the Navier–Stokes Equations in Self-similar Variables*, manuscript in this series, `docs/source/type_1/paper2_tight_ancient_stationary_rigidity.tex`.

[4] G. Duran Ballester, *Minimal Ancient Solutions for the Navier–Stokes Equations in Self-similar Variables*, manuscript in this series, `docs/source/type_1/paper3_minimal_ancient_solutions_reduction.tex`.

[5] G. Duran Ballester, *Compact Ancient Navier–Stokes Dynamics and Invariant Measures in Self-similar Variables*, manuscript in this series, `docs/source/type_1/paper4_compact_ancient_dynamics_invariant_measures.tex`.

[6] J. Nečas, M. Růžička, and V. Šverák, *On Leray's self-similar solutions of the Navier–Stokes equations*, Acta Math. **176** (1996), no. 2, 283–294, doi:10.1007/BF02551584.